Functional Characterisation of Polymer Electrolyte Fuel Cells: Novel In Situ Experimental Technique for Spatiotemporally Resolved Temperature Distributions

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Abstract
Fuel cell phenomena involves complex interactions of mass transport, energy transfer and electrochemical kinetics, which leads to spatial heterogeneity—ultimately affecting the performance. The direct measurement of this phenomena inside an operating fuel cell is a challenging task. Amongst all the available measurement techniques, local temperature measurements provide high resolution insights into all parts of an operating fuel cell. This work presents a novel in situ experimental technique to monitor temperature distribution along flow field (in plane) and across membrane electrode assembly (through plane) of a real scale fuel cell. A high spatiotemporal resolution localized temperature evolution that is due to reaction only was achieved. Thus, technique enables us to pinpoint the location and transient nature of hot spots. Temperature hot spots are one of the main reasons for fuel cell's degradation and eventual failure. This information can now be used to correlate other electrochemical parameters and material properties as well as to help improve cell design.

1. Introduction

Fuel cell converts chemical energy stored in the fundamental element directly into electricity and water by electrochemical reaction. As a consequence of this exothermic reaction useful heat is generated. In hostile operating conditions, the uncontrolled release of heat results into temperature hot spots that severely affects the cell performance, causes material degradation and its eventual failure. Temperature is one of the parameters with the highest level of spatial heterogeneity and temporal uncertainty as it is a function of internal operating conditions and depends on external factor such as flow rates, ambient conditions and feedback control. Temperature is directly or indirectly linked to (a) chemical factors: pressure, humidity, velocity, reactant composition, species diffusivities, and reaction rates; (b) electrical factors: current density, potential and contact resistance; (c) material properties: catalyst loading, decay, and membrane conductivity; (d) design: channel geometry and cell assembly. Therefore it is important to acquire—accurately—the localized absolute temperature measurements in all parts of the fuel cell. This information when combined with other functional parameters produces a functional map that can provide a complete understanding of its inner workings. Such a functional map can be used for the effective management of heat, electricity and water. In addition it can help in improving performance and cell design. However, in-situ temperature measurement using prior techniques have not been appropriately accounted for the occurrence of hot spots—when and where do they occur?

1.1 Impact of Thermal Stress and Heterogeneity

[*] *The financial support from EPSRC grants (EP/I014640/1 and EP/K503381/1) and sponsorships from CPSE industrial consortium is gratefully acknowledged.*

At elevated temperatures as the membrane dries out the internal resistance of the cell increases. The increase in internal resistance produces ohmic heating which develops into 'hot spots'. Other reasons for hot spots are poor distribution of fuel gases, high concentration of fuel gas at localized areas, channel blockage due to product formation or impurities, runaway reaction at some areas of the membrane electrode assembly (MEA) and localized shorting of the cell. This leads to material degradation, performance deterioration and ultimately cascading failure of the system: when a single cell fails, then the entire stack fails. Kandlikar and Lu, [2], have reviewed the water and thermal management issues while Wu et al., [4], have reviewed the available diagnostic tools and techniques highlighting various degradation modes. Recently Lin et al., [[] measured temperature distribution on the back side of the flow field plates to understand non-uniform temperature distributions. Their results were based on sensors placed at the surface of five serpentine channel flow field block with mathematical correlation to compute temperature on the surface of MEA. Due to large number of channels, imprecise measurement locations and low spatial resolution an accurate dynamic relationship between reaction rate and evolution of localized temperature distributions cannot be established with this technique.

From all the previously published results and technical reviews on thermal management issues we know that why and how hot spots occur in a unit cell. [Thiago/Anthony to add more details here]. But, the mysterious question still remains unanswered—when and where exactly do they occur? The published theoretical and experimental results do not pinpoint the location of temperature fronts or confidently ascertain or predict when and where the hot spots occur under different operational scenarios. By developing a novel in situ experimental technique this work gives the answers that can provide unprecedented insights into the inner working of a fuel cell. Additionally, the presented technique allows us to measure temperature rise due to reaction only and thus can be used to benchmark/compare performance of different types of fuel cells under real world operating conditions.

Table 1. Evolution of in situ thermal mapping techniques

Year	Authors	Remarks
2003	Mench et al., [165]	Embedded 8 micro-thermocouples directly between two 25 cm^2 thick Nafion electrolyte but only 3 gave reliable output. Cell tested with natural convective cooling and a 15 C temperature increase in the electrolyte is observed for current densities of 1 A/cm^2.
2004	Vie et al., [243]	Placed 4 micro-thermocouples in through-plane direction and found that anode side was higher than cathode.
2006	Wilkinson et al., [251, 250]	Embedded 19 micro-thermocouples on both sides bipolar plates with 280 cm^2 area; only 10 actually worked of which 9 on anode side. Effects of changes in current distribution in bipolar plate was ignored that affect accuracy due to electromagnetic interference.
2009	Lee et al., [134]	Method is improvement to [243] by inserting 7 thermocouples between layers in the through-plane direction without degradation of cell performance.
2010	Zhang et al., [263]	Simultaneous measurement of current and temperature distributions was obtained along 16 cm^2 serpentine channel flow field.
2011	Jiao et al., [108]	Simultaneously measured current and temperature distributions during various cold start processes using 16 thermocouples inserted into the 40 cm^2 3 channel cathode flow

		field.
2012	Lin et al., [135]	Placed an array of 25 thermocouples at the back sides of 50 cm^2 6 serpentine channels cathode and anode flow field plates. Due to sensor location and the high operating current densities true reaction temperature at MEA cannot be observed.

2. Methodology

Many methods are used to measure temperature distribution in a fuel cell. These methods include in situ measurements using micro sensors and ex situ imaging techniques, {e.g citeUCL}. Embedding micro sensors directly into the fuel cell renders inaccurate measurements due to increase in contact resistance and leakage. Additionally, their fabrication is a complex and expensive process. In contrast standard easy to use thermocouples based measurement is a preferred choice. Ex situ imaging based techniques fail to measure accurately the cathodic temperature distributions because of accumulation of liquid water in the channel flow field. Amongst all the measurement techniques, local temperature measurements with thermocouples are easier to set up. They can provide high resolution in situ spatial and temporal insights in an unobtrusive way with minimal impact on the fuel cell, \cite{wilkinson2006}. Pioneering work in this area was carried out by Mench et al., [3]. Table 1 presents number of in situ techniques that are being used to study fuel cells. Note that there are other in situ parameter measurement techniques include mapping of current density, \cite{Anthony}, electrolyte conductivity \cite{??}, water flooding \cite{neutronImaging}, as well as mapping the flow of reactant in gas channels being carried out, \cite{ThiagoPaper}. The combination of these techniques, applied across a range of fuel cell operating conditions, allows a unique picture of the internal workings of fuel cells and can be used to validate both numerical and analytical models, [1], \cite{AmitModellingPaper}. This work focuses on the accurate mapping of localized temperature distributions with highest possible spatial and temporal resolution. This information can be combined with the current distribution, \cite{Anthony}, flow/Pressure \cite{Thiago}, ionic conductivity, \cite{Cooper2011}.

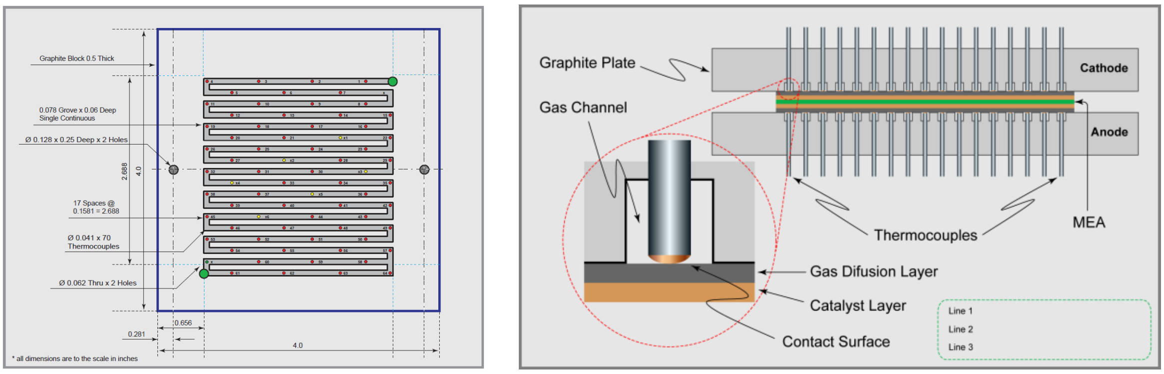

Figure 1. (a) Single serpentine flow field with thermocouple locations (b) Schematic of contact surface

2.1 Measurement Locations

To accurately monitor and detect temperature changes as they happen at the electrode, it is necessary to place the sensor as close to the electrodes (or electrolyte) as possible. The ideal location for placing the sensor probe is the contact surface with electrode where the reaction takes place. Mench et al., [165], presented a method for measuring the temperature distribution using micro-thermocouples placed inside the MEA between two sheets of Nafion. This method is difficult, intrusive as the diameter of the thermocouple is same as the thickness of the electrolyte (50 μm). Wilkinson et al., [251] approached differently by placing thermocouples in the landing area of the flow field plates. This is a better way of placing thermocouples in contact with the gas diffusion layer (GDL) and can be done in a noninvasive way to take in situ measurements. Lin et al., [135], used the same approach by placing thermocouples on the land area close to the surface of the flow field and used material conductivity correlation to map the temperature on the MEA surface. We have approached by placing array of thermocouples with highest possible spatial resolution to discern temperature between two neighboring probes by placing them directly on the surface of MEA in an unobtrusive way. This method presents the best approach for the in situ measurement as we are able to measure temperature on the surface of MEA that is due to reaction and not other effects. The schematic representation of our sensor array geometry on a 50 cm^2 single serpentine channel fuel cell is shown in Figure 1. This fully automated technique uses sensors on both anode and cathode side in exact opposite directions of the MEA as shown in Figure 2.

The accurate measurements of the absolute temperature distributions in real-time are described in the next section. The details of well-understood procedures and components are excluded to avoid unnecessary explanation. Albeit, the succeeding section describes results in detail to illustrate the embodiments of our proposed technique.

3. Experimental

3.1 Cell Design

A single pass serpentine channel design with dimensions: total channel length 52.7” long, the width 0.078” and the depth 0.062” shown in Figure 1 was machined on a 0.5’’ thick resin impregnated POCO industrial grade AXF-5Q graphite block (Fuel Cell Technologies). Single serpentine channel was chosen so as to measure localized heat released due to reaction only in a single channel. The pattern has 18 channels that are equally spaced with last two groves in parallel. An array of 70 thermocouples were places on both anode and cathode side with 10 mm spatial resolution. Type ‘E’ Thermocouple with dimensions 0.040” (1 mm) diameter and 6” long was chosen since it has the highest millivolt output for each degree of temperature. Thermocouples were purchased from Omega Engineering. Thermocouples are arranged such that they maintain contact with GDL surface as shown in Figure 2. The two thermocouples at locations marked ‘x’ were removed that were too close to the gas inlet/outlet. This arrangement renders the highest possible spatial resolution to distinguish temperature difference between neighboring thermocouples. The sensors were held by just the friction of the end plate with the Teflon tubing and an ‘O’ ring in the current collector to seal them. The 0.078” thick current collector plates were made of copper coated with gold. The fuel cell was assembled by clamping two anodized aluminium blocks using eight M8 screws each applied with uniform torque of 2 Nm. The conceptual design and the actual cell construction is presented in Figure 3 and 4.

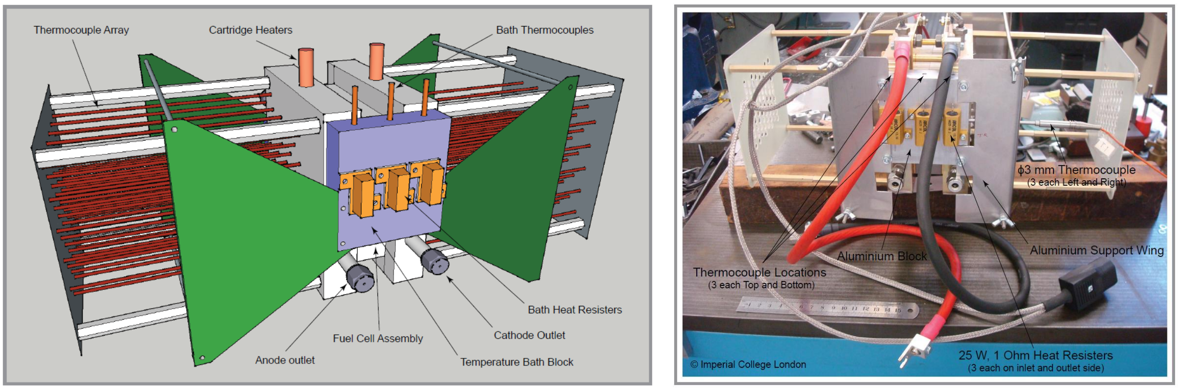

Figure 2. Design of in-house built experimental fuel cell (a) conception (b) fabricated unit

3.1 Thermal Bath

There are many factors that contribute to the non-uniformity of temperature distributions as discussed earlier. Since one of the major goals of this experiment is to measure the temperature distributions due to reaction only, the cell was designed to maintain a uniform thermal bath using an aluminium block with heat resisters mounted as shown in Figure 3 and 4. Bath controller shown in Figure 7 was designed in-house and can be powered by either traditional PID algorithm or advanced multi-parametric model predictive control algorithm, \cite{Manthanwar 2015}. With the combination of cartridge heaters and thermal bath the cell was maintained uniformly at 80 C. The experiment on test cell were conducted in controlled lab environment with constant temperature and humidity.

3.2 MEA Preparation

A seven layers MEA was utilized in this work, or two gas diffusion layers (carbon paper), two microporous layers, two catalyst layers and a Nafion™ membrane. Both anode and cathode electrodes were a commercial Johnson Matthey platinum electrode, at a platinum loading of 0.4 mg_{Pt} cm^{-2}, purchased from Alfa Aesar. The Nafion™ membrane was the N212, which has a thickness of 0.002”. This membrane was pretreated sequentially in 3 vol.% H_2O_2, de-ionized water, 0.5 M H_2SO_4, and D.I. water at 80 C for 1 h per step and kept in D.I. water prior to use. The MEA was obtained by hot-pressing a sandwich of electrodes and membrane at 135 ^{0}C with a pressure of 50 kg/cm^2 for 3 min.

3.3 Test Procedure

Figure 5 illustrates the components of the fuel cell used. Cell was connected to the multi range fuel cell test system (Model 850e, Scribner Associates Inc.) and computer controlled using the Scribner software, \cite{ Scribner2006}. The test rig is supplied with hydrogen gas (99.999%, BOC) to the anode and oxygen gas (99.997%, BOC) to the cathode. Both the gases were humidified with deionised water. The flow rate, humidity, temperature and pressure of the gases supplied were computer-controlled using Scribner software. The load was applied using constant voltage (potentiostatic) and constant current (galvanostatic) modes. Tests were carried out, while recording internal localized temperatures, under steady state and dynamic conditions by varying (a) voltage from open circuit voltage (OCV) to maximum current; (b) by changing stoichiometric flow rates at cathode to study starvation dynamics; and (c) by changing anode humidity. For steady state measurements sufficient time was allowed for the fuel cell to reach a steady state operating condition.

Once this steady state condition was reached, temperature data was recorded for a constant period of time and later samples were averaged. A set of controlled experiments were performed in an ideal conditions.

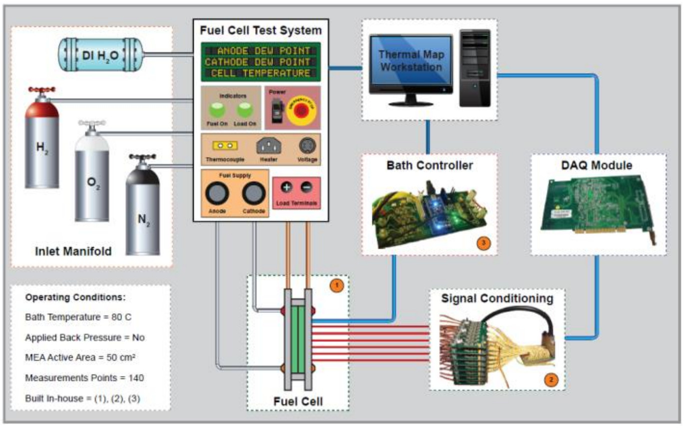

Figure 3. Experimental test setup with in-house built components 1, 2, and 3

3.3 Thermocouple Calibration

Thermocouples were calibrated to give 0.25 C resolution with 90% confidence. The calibration curve is presented in Figure 7 and the dynamic response of the thermocouples from 20 to 60 C is presented in Figure 8. The result of static and dynamic calibration analysis is presented in the Table 2.

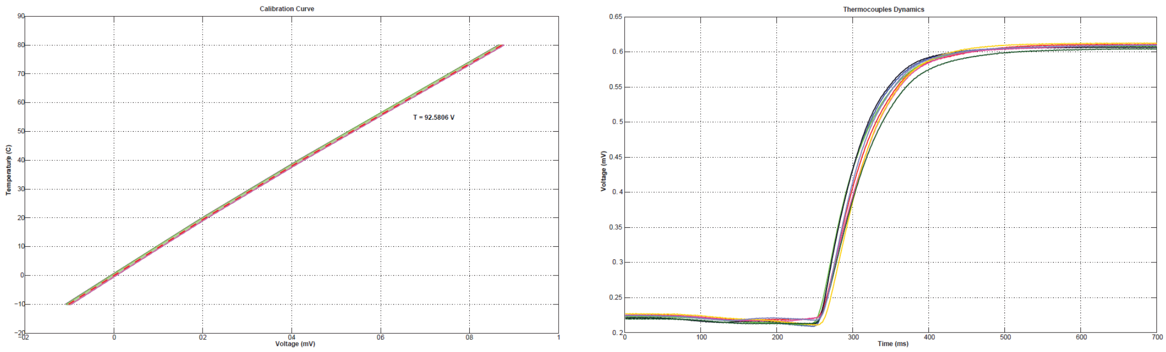

Figure 4. Thermocouple static calibration curve and dynamic characteristics

Table 2. Thermocouple Characterization

Accuracy with 95 % Confidence	0.3 C
Accuracy with 100% Confidence	0.5 C
Resolution	0.5 C
Offset at 0 C	(-) 0.0394 mV
Minimum Reading	(-) 10 C
Maximum Reading	80 C
Dynamic Settling Time	300 ms
Dynamic Time Constant	75 ms
Response Time	100 ms

4. Results and Discussion

4.1 Polarization curves

Figure 5 depicts the polarization curve with maximum power density of 641 mW/cm^2 at 4.4 V. At 0.8 V the power density is about 400 mW/cm^2. When the current density is at 1000 mA/cm2, the voltage is close to 0.6 V. The inner and outer rectangle in the Figure 5 describes the region of safe and extreme operation zones. We can observe that the cell has a fairly large operating window of current density between 300 to 1100 mA/cm^2 and voltage between 0.55 to 0.75 V for safe operation. Operation beyond 1400 mA/cm^2 causes significant drop in power density and voltage values due to temperature rise that is undesirable. This temperature rise can be seen from the dynamic temperature profiles presented in the subsequent section to follow.

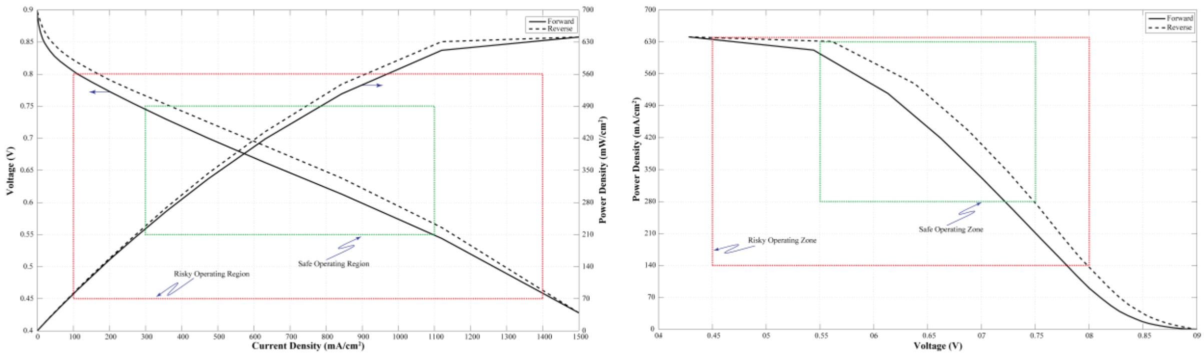

Figure 5. Performance characteristics for 1.5 λ_{H2} and 2.5 λ_{O2}

4.2 Dynamic Temperature Profiles

Controlled experiments performed in ideal conditions provide insight into operational cause-and-effect behaviour of a fuel cell by observing the outcome when a particular variable or combination of variables are manipulated. The repeatable outcome and logical analysis of results help advance the understanding of a fuel cell phenomenon. The ‘step test’ or step response analysis is one the best ways to study the dynamic behavior of a fuel cell system output for a sudden sufficiently large orthogonal step change in system input (manipulated variable) while keeping all the other system variable constants. Such a carefully designed test if carried out in an ideal conditions can help estimate system’s time response to reach new steady state (settling time), time constant (space time) which roughly tells how much time system takes to reach (1-1/e ~ 63.21% if increasing or 1/e ~ 36.29% otherwise) of final steady state value, time delay, and the impact of internal system parameters on the stability of output. Data collected using step test can be subsequently used to validate system’s dynamic mathematical model. Figure 6 shows the envelope of transient temperature evolution for the incremental step change in voltage as staircase from 0.8 V to 0.4 V with 0.1 V increment.

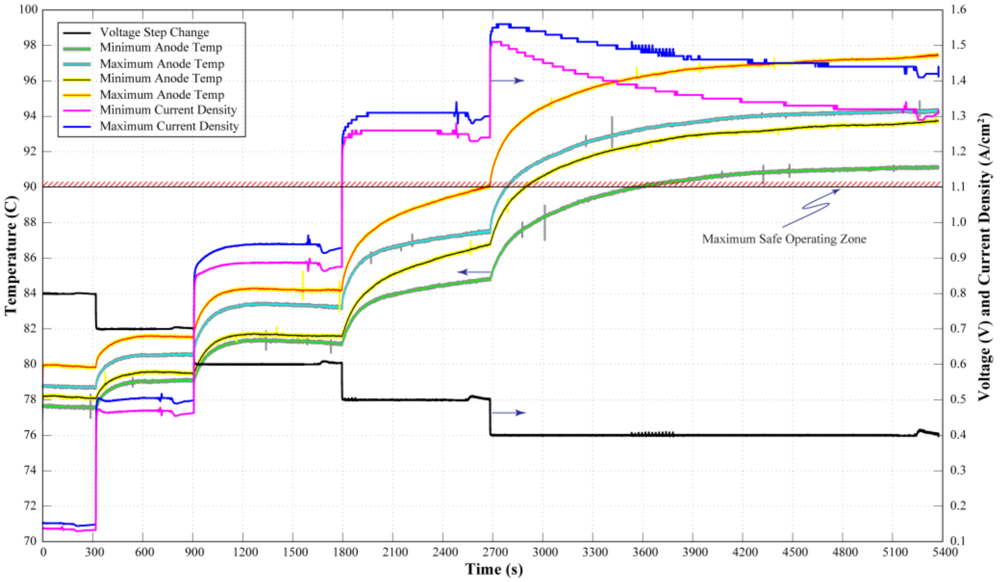

Figure 6. Dynamic evolution of temperature and current density for voltage step test

The step length was kept for sufficiently long duration for the temperature response to reach a new steady state. From the plot we can observe that at 0.5 V the anode temperature is close to the maximum allowable operating temperature of 90 C. At 0.4 V it quickly crosses that safety bound and from the current density response we can notice that the performance of the cell begins to deteriorate. This deterioration is due to the fact that as the cell dries out the activity decreases, yet the temperature continues to increase due to ohmic heating. This is the point when hot spots begins to appear which we can be seen from the in-plane view of the snapshot taken at steady state shown in Figure 7 below.

From 0.8 V to 0.5 V we can observe that the temperature rise is almost linearly proportional to current density. When cell is maintained below 0.5 V, Current reaches beyond 1200 mA/cm2 and temperature sharply shoots up over 10 C as observed in Figure 6 and 7. From the Figure 7(c) we can observe that the power starts to drop and cell starts to deteriorate its performance.

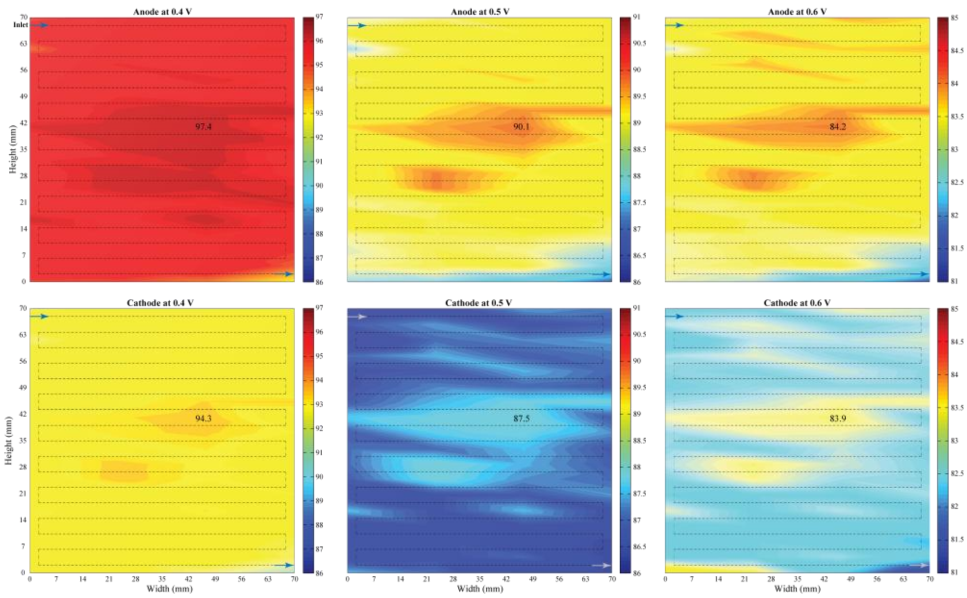

Figure 7. Steady state temperature distributions at constant voltage

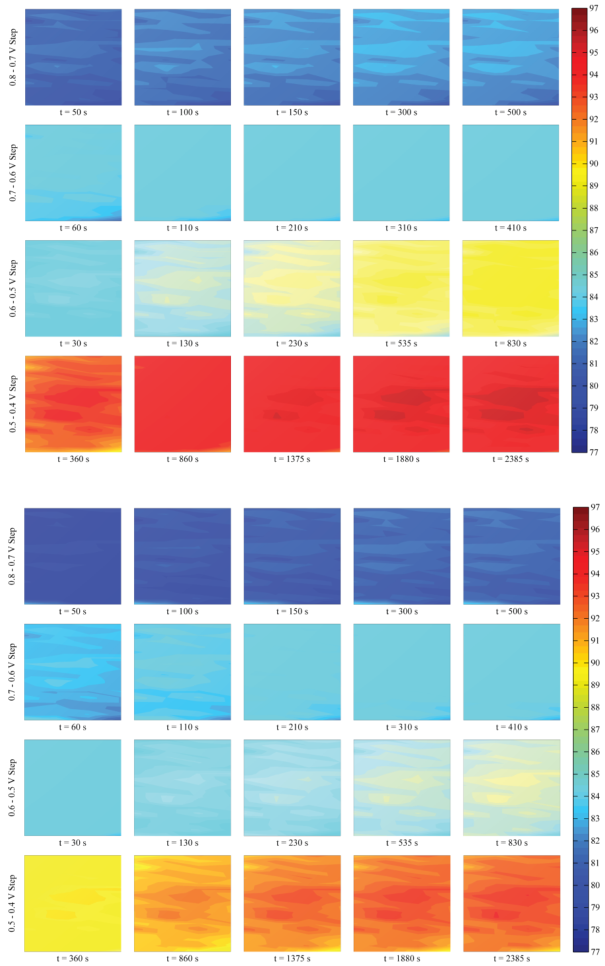

Figure 8. Snapshot of dynamic temperature map on left anode and right cathode side

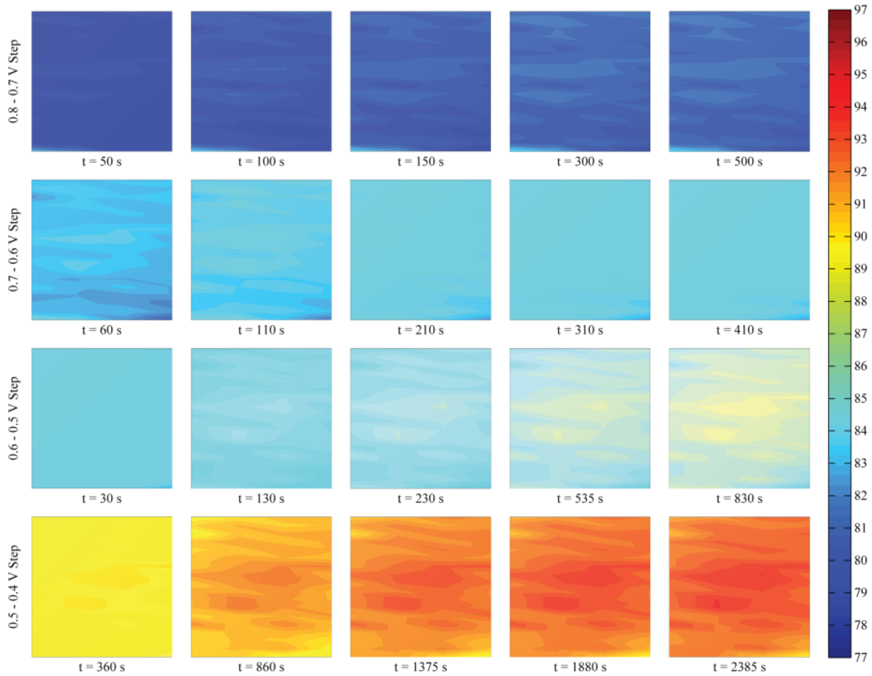

Figure 9. Snapshots of temperature distribution on cathode side

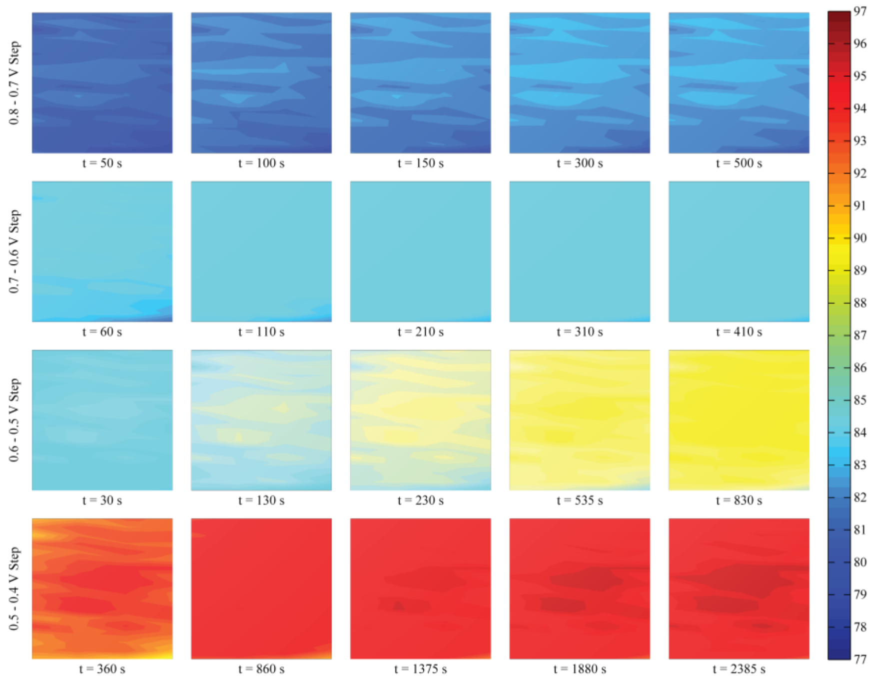

Figure 10. Snapshots of temperature distribution on cathode side

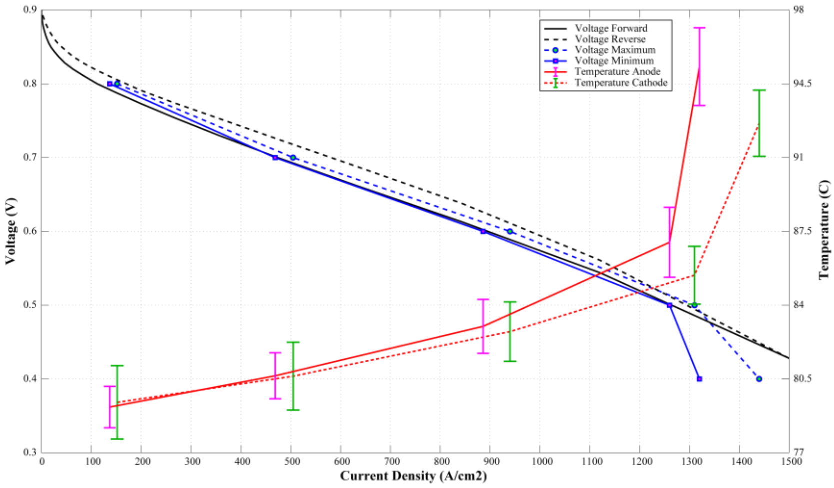

Figure 11. Voltage-current-temperature characteristics

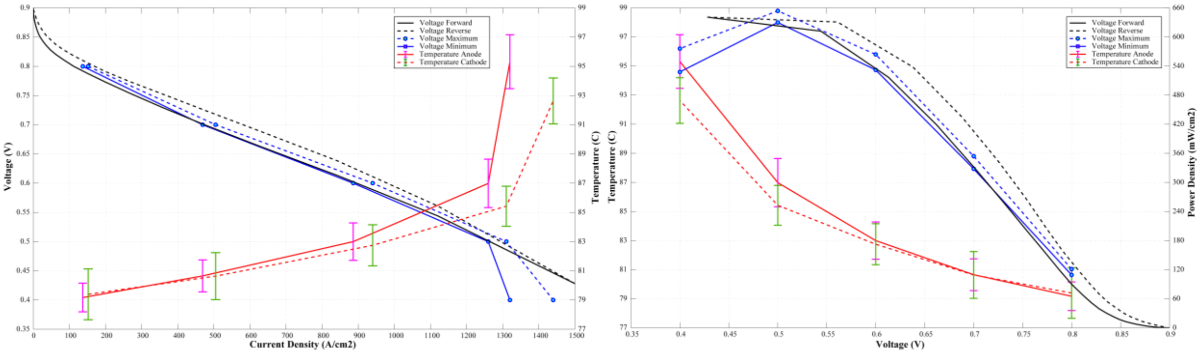

Figure 12. Voltage-current-temperature characteristics

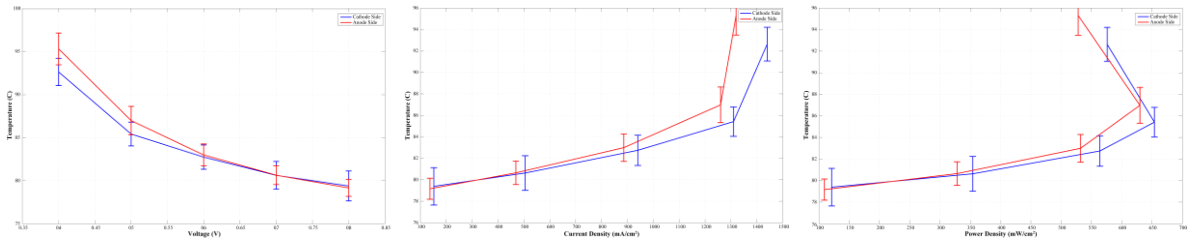

Figure 13. Mean anode and cathode side temperature characteristics

4.3 Steady State Temperature Distribution Along the Flow Channel

Controlled experiments performed in ideal conditions provide insight into operational cause-and-effect

4.4 Temperature Distribution Across the MEA

Controlled experiments performed in ideal conditions provide insight into operational cause-and-effect

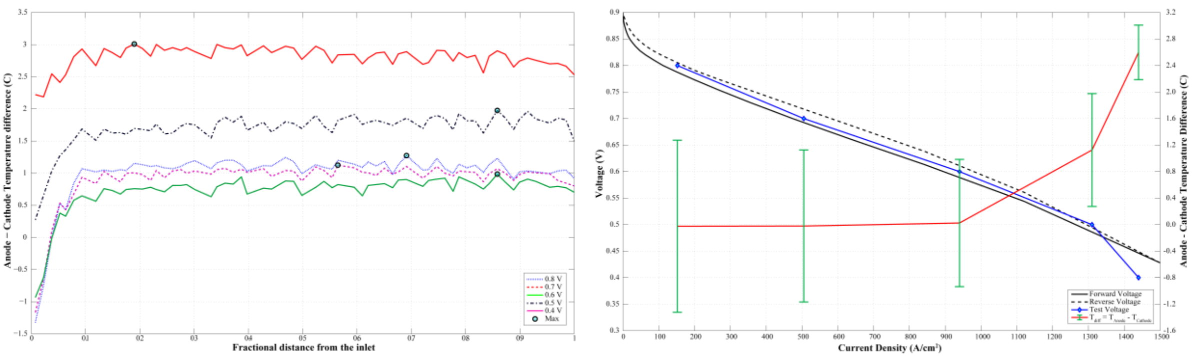

Figure 14. Characteristics of anodic and cathodic temperature difference

4.5 Temperature Distribution Along the Flow Channel

Controlled experiments performed in ideal conditions provide insight into operational cause-and-effect

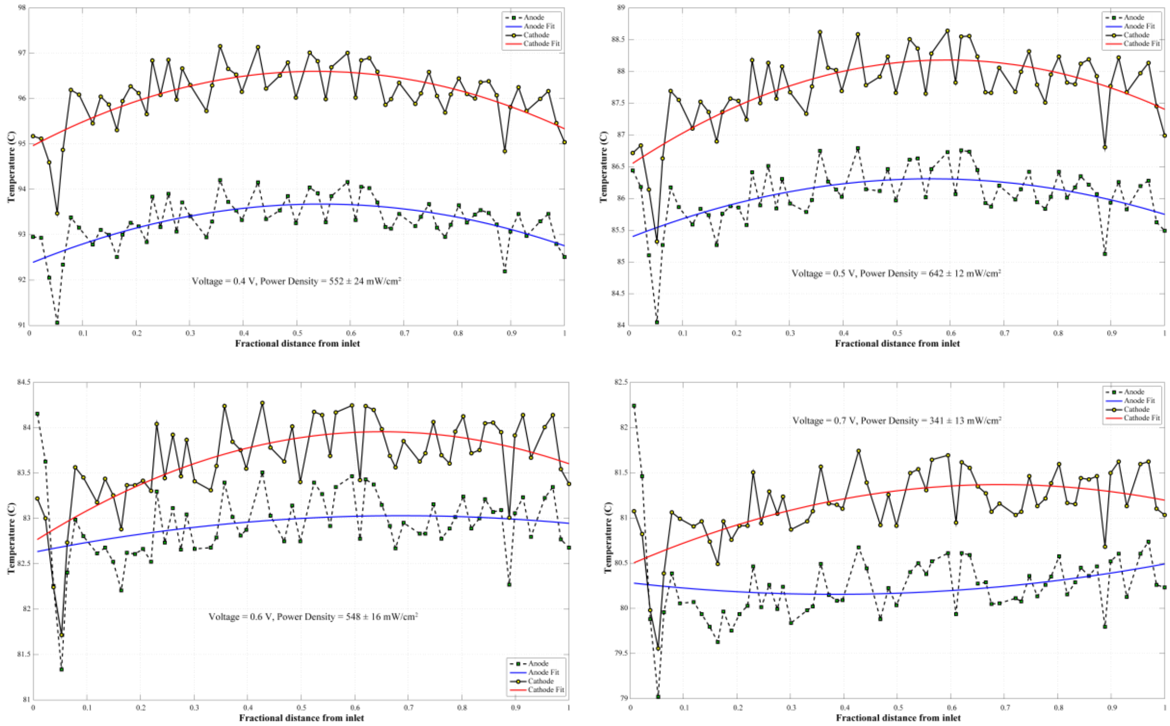

Figure 15. In plane direction temperature distribution at steady state operation

4.6 Impact of Uncertainty on Temperature Distribution

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4.7 Impact of Oxygen Starvation on Temperature Distribution

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4.8 Thermal Map Movie Along the Surface

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Conclusion

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At elevated temperatures as the membrane dries out the internal resistance of the cell increases and the power output significantly reduces.

Temperature measurement due to reaction only correlates well with other electrochemical parameters and material properties.

, thus now able to answer long standing questions why, how, when and where the hot-spots occur.

These spatially and temporally resolved measurements provide an insight into operational cause-and-effect behaviour of fuel cells by observing the outcome when a particular variable or combination of variables are manipulated.

Possible degradation mechanisms and fuel cell failure modes due to occurrences of hot-spots are well documented in open literature.

This method is an effective tool to test existing theory and critically examine the inner workings of a fuel cell accurately thereby help deduce possible degradation mechanisms and eventual failure modes due to occurrences of hot-spots.

The repeatable outcome and logical analysis of results help advance the understanding of fuel cell phenomenon.

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thereby help deduce a new mathematical model. We placed 64 thermocouples on both anode and cathode side in exact opposite directions of the membrane electrode assembly touching the gas diffusion layers. A set of controlled experiments were performed in ideal conditions by varying humidity, fuel stoichiometry and power output. These spatially and temporally resolved measurements provide an insight into operational cause-and-effect behaviour of fuel cells by observing the outcome when a particular variable or combination of variables are manipulated. The repeatable outcome and logical analysis of results help advance the understanding of fuel cell phenomenon.

Several factors critical to the cell performance include electrochemical parameters, material properties and design.

Only spatiotemporally resolved measurements can accurately provide insight into all parts of an operating fuel cell.

This method is an effective tool to test existing theory and critically examine the inner workings of a fuel cell accurately.
We present a novel experimental method for understanding the temperature distribution across (through plane) membrane electrode assembly and along (in plane) flow field of a 50 cm2 single serpentine channel fuel cell. This method is an effective tool to test existing theory and critically examine the inner workings of a fuel cell accurately thereby help deduce a new mathematical model. We placed 64 thermocouples on both anode and cathode side in exact opposite directions of the membrane electrode assembly touching the gas diffusion layers. A set of controlled experiments were performed in ideal conditions by varying humidity, fuel stoichiometry and power output. These spatially and temporally resolved measurements provide an insight into operational cause-and-effect behaviour of fuel cells by observing the outcome when a particular variable or combination of variables are manipulated. The repeatable outcome and logical analysis of results help advance the understanding of fuel cell phenomenon.

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5. Appendices

5.1 Correlation for Temperature Dependent Current Density

Simultaneous measurement of sporadic local temperature distribution with current density distribution is a challenging task. One of the simpler way to relate dynamic temperature distributions with current density used by Wilkinson et al., [251] as follows,

$$i_p = \frac{mC_p}{(V_0 - V)} \frac{dT}{dt} \tag{3}$$

where m is the thermal mass surrounding the thermocouple (in kg), C_p is the specific heat capacity of the thermal mass surrounding the thermocouple (in kJ/kg·K), T is the local temperature (in K).

More importantly when developed it can answer the most important questions with regard to the temperature variations in fuel cells and

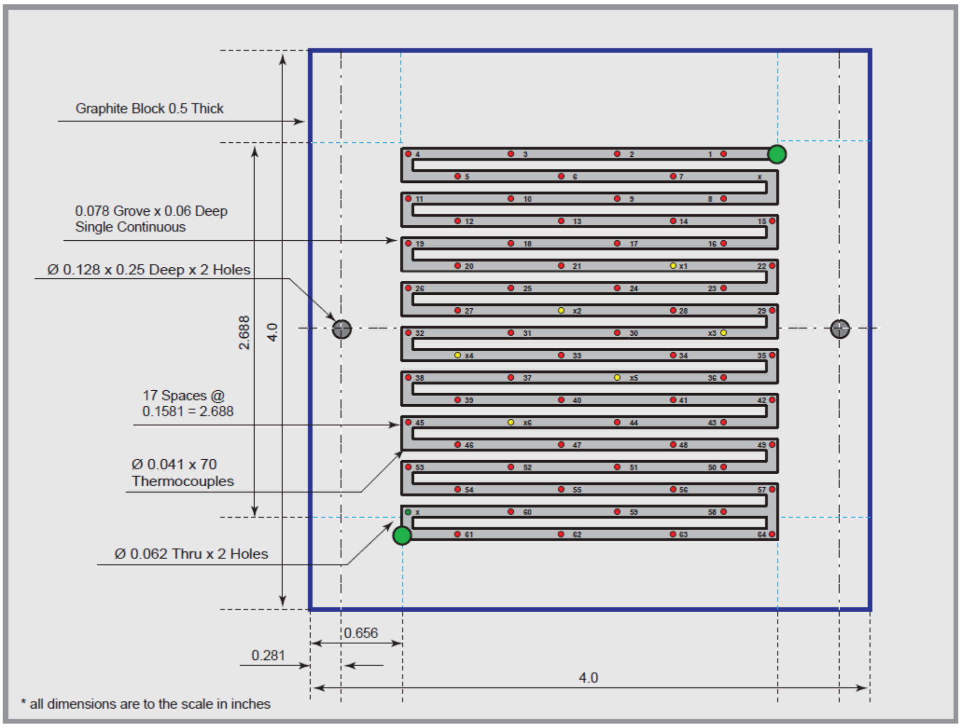

Figure 16. Single serpentine flow field on graphite block with thermocouple locations

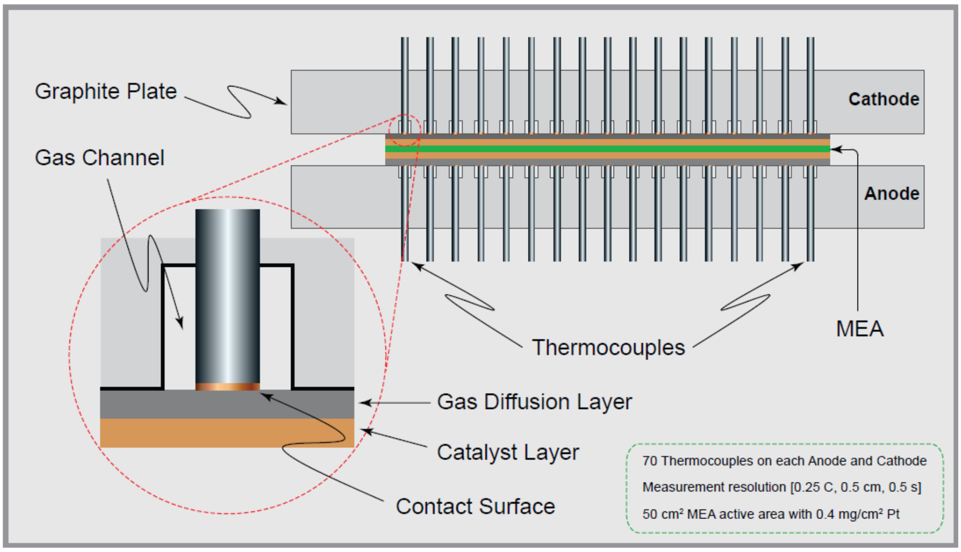

Figure 17. Schematic of contact surface of temperature measurement locations

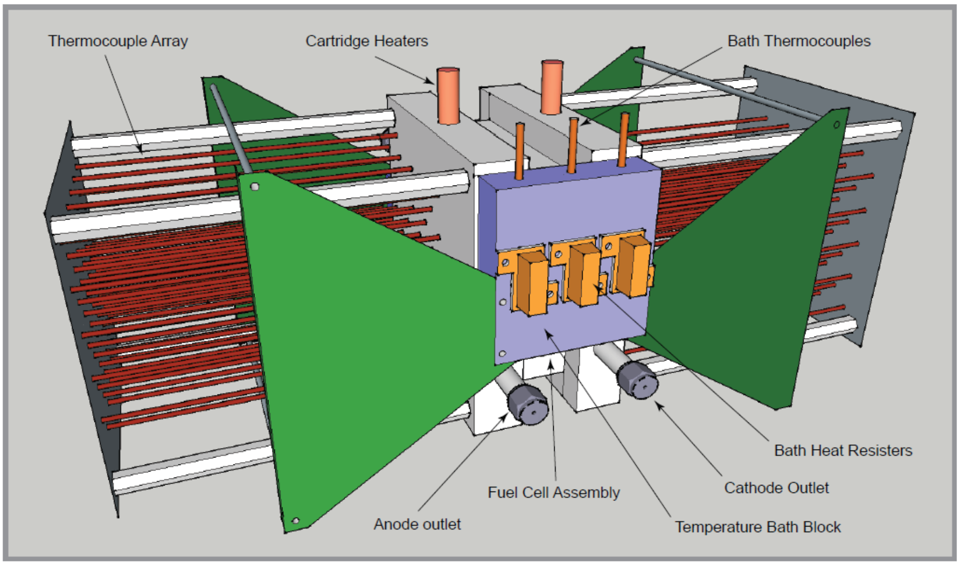

Figure 18. Design of in-house built experimental fuel cell

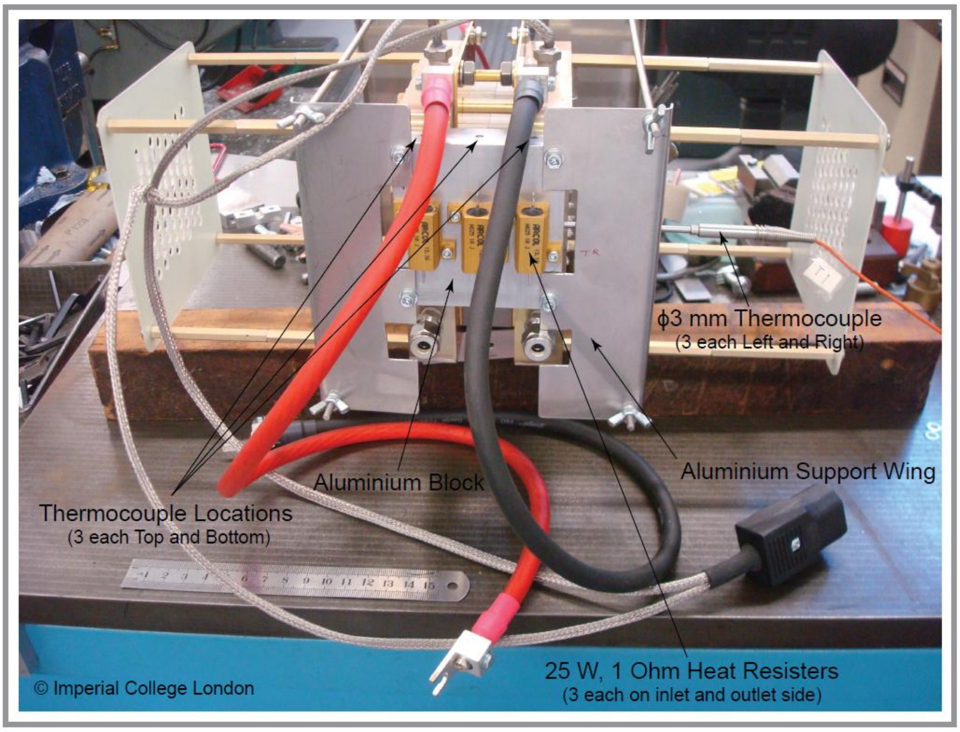

Figure 19. Fully constructed experimental fuel cell with thermal bath blocks

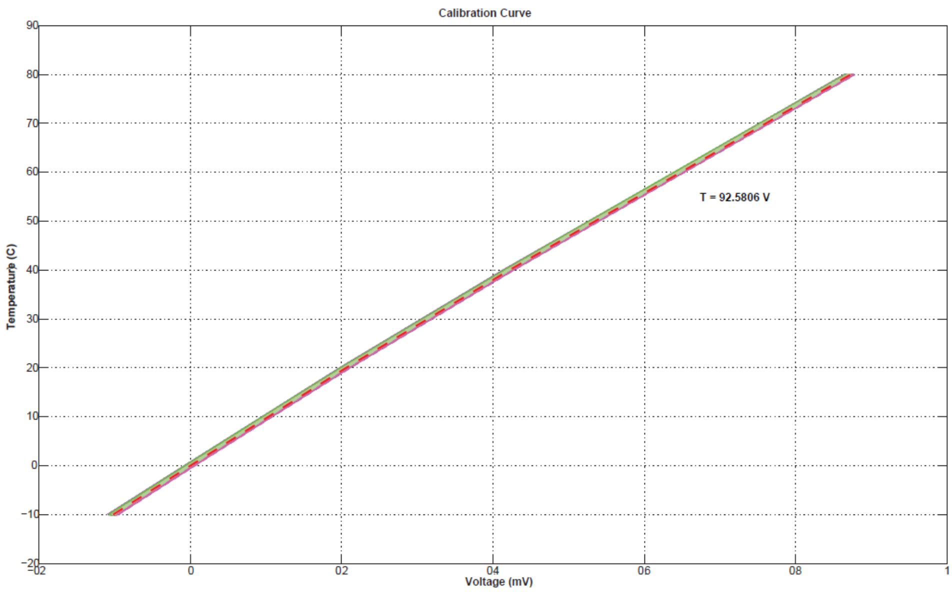

Figure 20. Thermocouple static calibration curve

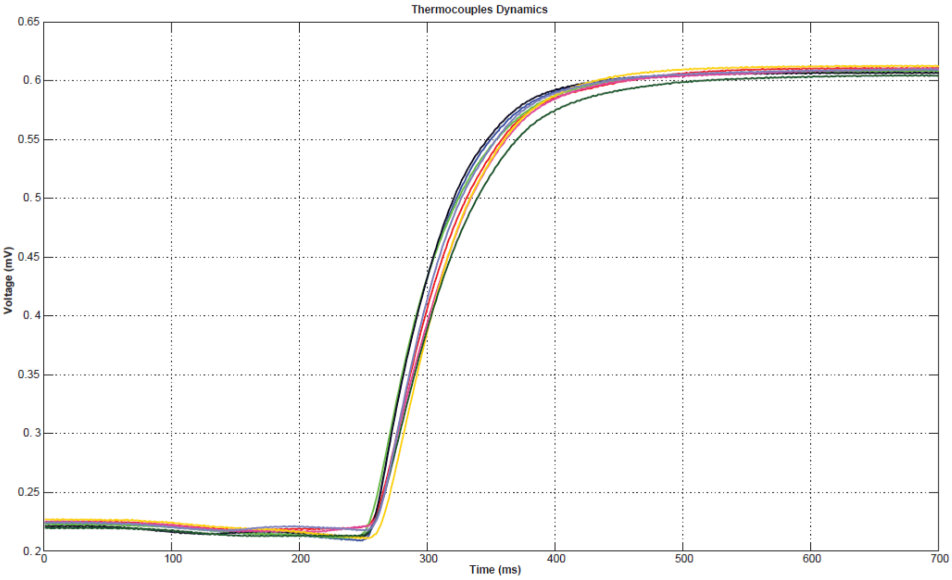

Figure 21. Dynamic characterisation of thermocouples